

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CE) Examination

Nov-Dec 2013

CE 207 Java Programming

Date: 12.11.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following questions.****[07]****State True or False.**

- 1) Java technology is both a programming language and a platform.
- 2) A .class file contains byte codes.
- 3) The modifiers public and static cannot written in either order "public static" or "static public".
- 4) Constructor overloading is not possible in Java.

Fill in the Blanks.

- 5) Another term for data hiding is _____.
- 6) _____ means that two or more methods can have different names in the same way that an English word can have two or more meanings.
- 7) The parameter to main is a (n) _____.

Q - 2 Answer the following questions.**(A) What is package? Write steps to create package with example****[07]****(B) Write the differences between interface and abstract class. Is it possible to have reference variable of type interface or abstract class? Justify your answer.****[07]****OR****Q - 2 Answer the following questions.****(A) List the types of java operator. Explain any one of them briefly.****[07]****(B) Write the difference between Exception and Error. Explain how exceptions are handled in program.****[07]****Q - 3 Answer the following questions. (with output)****(A) Write an application that declares a class named Sphere. It should have instance variable to record its radius and the coordinates of its center. These should be type double. Use the new operator to create a Sphere object. Set and display its instance variables.****[05]****(B) A trapezoid is a geometric shape that has four sides. Two of these are parallel and are called the bases of the trapezoid. The height is the distance between the bases. Define a trapezoid class that encapsulates the height and the length of each base. Provide a constructor that initializes these instance variables. Provide one instance method that calculates the area of the shape. The area equals $0.5 * \text{height} * (\text{base 1} + \text{base 2})$. Instantiate an object of this class and invoke its method.****[05]****(C) Write an application that demonstrates a class inheritance hierarchy. Class M extends Object and has two instance variables of type float and String. Class N extends M and has one instance variable of type double. Instantiate class N. initialize and display its variable.****[04]**

OR

Q - 3 Answer the following questions. (with output)

- (A) Declare an abstract class vehicle with an abstract method named numWheels (). [05]
Provide subclasses Car and Truck that each implement this method. Create instances of these subclasses and demonstrate the use of this method.
- (B) Write a program that illustrates interface inheritance. Interface P is extended by P1 and P2. Interface P12 inherits from both P1 and P2. Each interface declares one constant and one method. Class Q implements P12. Instantiate Q and invoke each of its methods. Each method displays one of the constants. [05]
- (C) Write a program that prints the numbers 1 to 100 using five columns. Have each number separated from the next by a tab. [04]

SECTION - II

Q - 4 Answer the following questions.

[07]

Fill in the Blanks.

- 1) The reserved word _____ can be used as a name for the calling object.
- 2) _____ is the word used to say that there are no restrictions on the use of a method or data item can be used.
- 3) A (n) _____ is a special kind of method used to initialize objects.
- 4) An array can be initialized by enclosing a list of values in _____.

State True or False.

- 5) Constructors have return type void.
- 6) What is the value of the expression $(x > 3) \parallel (y < 3)$ if x has the value 1 and y has the value 5?
- 7) What will the following code print?

```
int color = 3;
switch (color)
{
    case 1:
    case 2:
        System.out.print("red ");
    case 3:
        System.out.print("blue");
        break;
    case 4:
        System.out.print("purple");
        break;
    default:
        System.out.print("gray");
}
```

Q - 5 Answer the following questions.

- (A) Explain String class. How StringBuffer differs from String class. [07]
- (B) Explain Following terms. [07]
Class, Object, inheritance, Polymorphism, method overloading, method overriding and Data Encapsulation

OR

Q - 5 Answer the following questions.

- (A) What is an applet? How it differs from application? Explain <APPLET> tag. [07]
- (B) What is Runtime Exception? Explain Unchecked and Checked exceptions. [07]

Q - 6 Answer the following questions. (with output)

- (A) Write an application that executes two threads. One thread displays "A" every 3000 milliseconds, and the other displays "B" every 5000 milliseconds. Create the threads by extending the Thread class. [05]
- (B) Write a multithreaded program that simulates a set of grasshoppers jumping around in a box. Each grasshopper jumps to a different location after every 2 to 12 seconds. Display the new location of a grasshopper after each of these jumps. [05]
- (C) Write an applet that draws a rectangle. The dimensions of the applet should be 800 * 400 pixels. (Use drawRect () method of graphics.) [04]

OR

Q - 6 Answer the following questions. (with output)

- (A) Write an application that executes two threads. One thread displays "A" every 2000 milliseconds, and the other displays "B" every 4000 milliseconds. Create the threads by implementing the Runnable interface. [05]
- (B) Write an applet that draws a circle. The dimensions of the applet should be 500 * 300 pixels. The circle should be centered in the applet and have a radius of 100 pixels. (Use drawOval () method of graphics.) [05]
- (C) Write a program that prints only the odd numbers between 1 and 100. [04]
